

Design and Analysis of a Bidirectional Shift Register using FSM

1 Introduction

A **Bidirectional Shift Register** is a sequential digital circuit capable of shifting stored binary data either to the left or to the right based on a control input. It consists of flip-flops connected in such a way that data can move in both directions.

It is modeled as a **Finite State Machine (FSM)** because:

- It has a finite number of states (2^n for an n -bit register)
- The next state depends on the current state and inputs

2 Components Required

- 4 D Flip-Flops
- Multiplexers (2:1) for direction control
- Clock signal (CLK)
- Direction control input (DIR)
- Serial input (SI)

3 Inputs and Outputs

Inputs:

- CLK: Clock signal
- DIR: Direction control (0 = Right Shift, 1 = Left Shift)
- SI: Serial input

Outputs:

- Q_3, Q_2, Q_1, Q_0 : Register contents

4 State Representation

Each state represents the current contents of the register:

$$(Q_3 \ Q_2 \ Q_1 \ Q_0)$$

Total number of states:

$$2^4 = 16$$

5 State Table

Present State	DIR	SI	Next State
$Q_3 Q_2 Q_1 Q_0$	0	SI	$SI \ Q_3 \ Q_2 \ Q_1$
$Q_3 Q_2 Q_1 Q_0$	1	SI	$Q_2 \ Q_1 \ Q_0 \ SI$

6 Next State Equations

Using D Flip-Flops:

$$D_3 = \overline{DIR} \cdot SI + DIR \cdot Q_2$$

$$D_2 = \overline{DIR} \cdot Q_3 + DIR \cdot Q_1$$

$$D_1 = \overline{DIR} \cdot Q_2 + DIR \cdot Q_0$$

$$D_0 = \overline{DIR} \cdot Q_1 + DIR \cdot SI$$

7 Working Principle

The operation of the bidirectional shift register depends on the value of the direction control input (DIR):

7.1 Right Shift (DIR = 0)

- Data shifts towards the right
- Serial input (SI) enters the leftmost flip-flop
- Example:

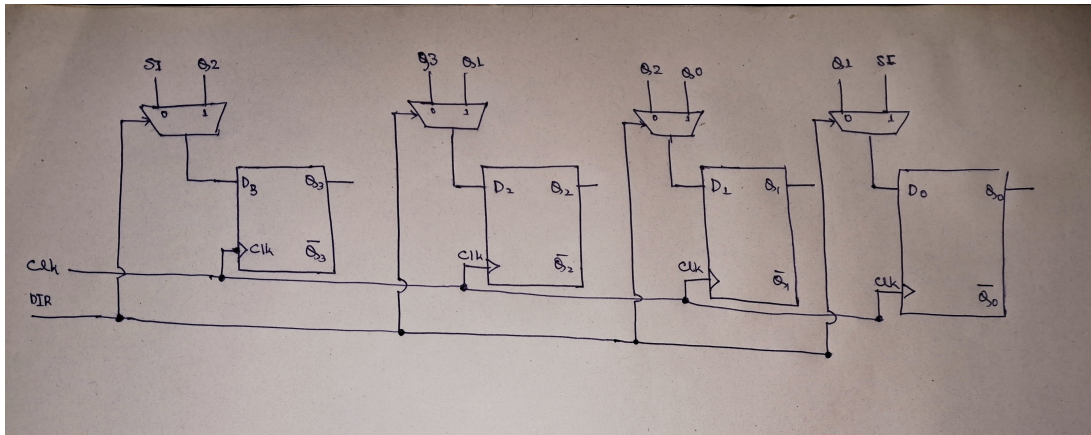
$$Q_3 Q_2 Q_1 Q_0 = 1011 \rightarrow 1101 \quad (SI = 1)$$

7.2 Left Shift (DIR = 1)

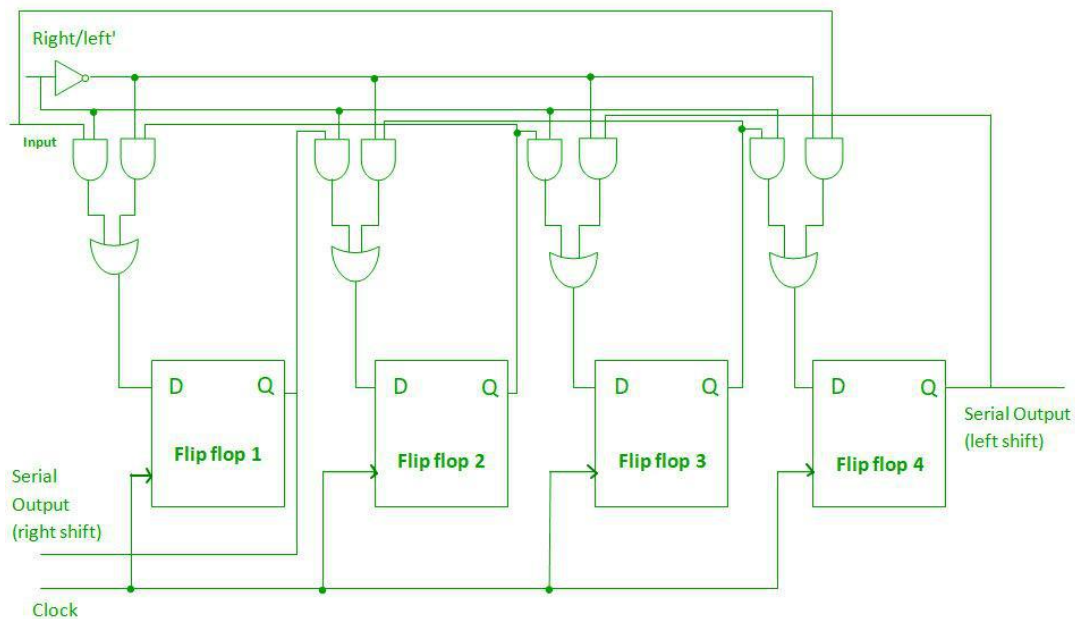
- Data shifts towards the left
- Serial input (SI) enters the rightmost flip-flop
- Example:

$$Q_3 Q_2 Q_1 Q_0 = 1011 \rightarrow 0110 \quad (SI = 0)$$

8 Circuit Diagram



9 Circuit Diagram (By Chip-GPT)



10 Applications

- Data storage and transfer
- Serial to parallel and parallel to serial conversion
- Arithmetic operations
- Digital communication systems

Verilog Implementation and Timing Analysis

Design Code

```
1 `timescale 1ns/1ps
2
3 module bidirectional_shift_register (
4     input clk,
5     input reset,
6     input dir,
7     input si,
8     output reg [3:0] q
9 );
10
11 always @(posedge clk or posedge reset) begin
12     if (reset)
13         q <= 4'b0000;
14     else if (dir == 1'b0)
15         q <= {si, q[3:1]}; // Right shift
16     else
17         q <= {q[2:0], si}; // Left shift
18 end
19
20 endmodule
```

Testbench Code

```
1 `timescale 1ns/1ps
2
3 module tb_bidirectional_shift_register;
4
5 reg clk;
6 reg reset;
7 reg dir;
8 reg si;
9 wire [3:0] q;
10
11 bidirectional_shift_register uut (
12     .clk(clk),
13     .reset(reset),
14     .dir(dir),
15     .si(si),
16     .q(q)
17 );
18
19 always #5 clk = ~clk;
```

```

20
21 initial begin
22     $dumpfile("dump.vcd");
23     $dumpvars(0, tb_bidirectional_shift_register);
24
25     clk = 0;
26     reset = 1;
27     dir = 0;
28     si = 0;
29
30     #10 reset = 0;
31
32     // Right shift
33     dir = 0; si = 1; #10;
34     si = 0; #10;
35     si = 1; #10;
36
37     // Left shift
38     dir = 1; si = 0; #10;
39     si = 1; #10;
40
41     #20 $finish;
42 end
43
44 initial begin
45     $monitor("Time=%0t | DIR=%b SI=%b Q=%b", $time, dir, si, q);
46 end
47
48 endmodule

```

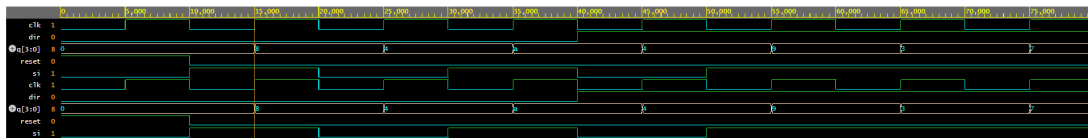
Code Explanation

- The module implements a 4-bit bidirectional shift register using behavioral modeling.
- The register `q[3:0]` represents the current state of the system.
- The `always @(posedge clk or posedge reset)` block ensures that:
 - Data shifts occur on the rising edge of the clock.
 - The register is asynchronously reset to 0000.
- When `dir = 0`, a right shift operation is performed:

$$q \leftarrow \{si, q[3:1]\}$$
- When `dir = 1`, a left shift operation is performed:

$$q \leftarrow \{q[2:0], si\}$$
- Non-blocking assignment (`<=`) is used to correctly model flip-flop behavior.

Timing Diagram



Explanation of Timing Diagram

- Initially, when `reset` = 1, the output is cleared to 0000.
- For `DIR` = 0, the register performs right shift:
 - Data shifts towards the right.
 - Serial input enters at the most significant bit.
 - Example:
$$0000 \rightarrow 1000 \rightarrow 0100$$
- For `DIR` = 1, the register performs left shift:
 - Data shifts towards the left.
 - Serial input enters at the least significant bit.
 - Example:
$$1010 \rightarrow 0100 \rightarrow 1001$$
- Output transitions occur only at clock edges, confirming synchronous operation.
- The waveform verifies correct FSM behavior, where each register value represents a valid state.